



Roll No:

2500020105012

MTECH
(SEM I) THEORY EXAMINATION 2025-26
FOUNDATION OF COMPUTER SCIENCE

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A1. Attempt *all* questions in brief.

02 x 7 = 14

Q no.	Question
a.	What is a Priority Queue? How is it different from a standard queue?
b.	Differentiate between Linear and Non-Linear data structures with examples.
c.	What are the necessary conditions for a Deadlock to occur?
d.	Explain the concept of Context Switching.
e.	Define Chomsky Normal Form (CNF).
f.	What are ACID properties?
g.	Define Hashing. What do you understand by the term 'Collision'?

SECTION B2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question
a.	Explain Merge Sort. Show the divide and conquer steps for {38, 27, 43, 3, 9, 82, 10}.
b.	Explain the concept of Multithreading. Differentiate between User-level threads and Kernel-level threads.
c.	Explain the closure properties of Regular Languages and Context-Free Languages.
d.	Explain Normalization. Discuss 1NF, 2NF, 3NF, and BCNF with suitable examples.
e.	Explain the Entity-Relationship (ER) Model. Draw an ER diagram for a "University Management System" considering entities like Student, Course, Department, and Faculty.

SECTION C3. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Explain Breadth First Search (BFS) and Depth First Search (DFS) traversal algorithms. Compare their complexities and applications.
b.	Define B-Trees. Construct a B-Tree of order 5 with the following data: {1, 12, 8, 2, 25, 5, 14, 28, 17, 7, 52, 16, 48, 68, 3, 26, 29, 53, 55, 45}.